Sharpening Handsaws

TEXT, PHOTOS, AND DRAWINGS BY PAUL SELLERS

Handsaws are indispensable to woodworking. A hundred years ago, the basics of maintaining and sharpening these saws were integrated into every woodworker's life, handed down through successive generations of craftsmen. Most woodworkers no longer practice these skills, but sharpening takes only a few practice sessions to learn, and incorporating these simple methods into your shop skills will enable you to get the most out of this essential tool.

Saws come in a wide range of shapes and sizes that may seem confusing at first, but it helps to remember that there are really only two types: ripsaws and crosscut saws. Though the sawing action is the same for both types, the basic structure of wood requires that the teeth be cut and filed in two distinctly different ways. The ripsaw uses chisel-type teeth to cut a groove along the grain. On the crosscut saw, pointed knife-like teeth shear the cross-grain fibers of the sidewalls on both sides of the cut (1).

Saws vary in size from massive, two-man saws for tree-cutting and dimensioning lumber to tiny saws for cutting small sections of wood. Saw teeth vary in size according to the size of wood you want to cut. Saw teeth are gauged by the number of teeth per inch (tpi) of saw length, or, more commonly, the number of points per inch (ppi) of saw length; this is often referred to simply as, for example, a 10-point saw. However, these two numbers are not the same. The number of teeth per inch is measured from gullet to gullet, whereas the number of points per inch is measured from tip to tip, so the tpi is always one less than the ppi. In other words, our 10-point saw will have 9 teeth per inch.

There are two ways to know if a saw needs sharpening. The obvious one is that it simply will not cut or it's difficult



Keeping your handsaws sharp is essential for accurate—and satisfying—use of this basic tool. With a little instruction and a few practice sessions, you can learn this important skill.

to cut with. The other way is to look at the points of the teeth. If the tips shine, the saw is dull. Light cannot reflect off a sharp edge.

SAW TOOTH SHAPE

Before we consider sharpening, however, we need to understand that each saw tooth has a pitch or rake angle that may be varied to suit your individual preference. The pitch is the sloping angle to the front, cutting edge of each saw tooth. It's important to maintain consistent pitch to the front of each tooth; otherwise, one part of the saw cuts more aggressively than the other.

Most manufactured saws come with a standard pitch of around 6°-8° for ripsaws and 13°-15° for crosscut saws. You may choose to have more pitch on one saw and less on another: one for finer, less aggressive cuts, and the other for coarser cuts (2). Many craftsmen file their ripsaws with a zero pitch (perpendicular to the length of the saw), which causes the saw to cut aggressively. You may even want to adjust the angle for one operation or a series of cuts and then revert back to some more general angle later. With practice you will gain the experience to alter the pitch of a saw in just a few minutes, but most craftsmen find an angle that suits them and stay with it.

VISES AND SAW FILES

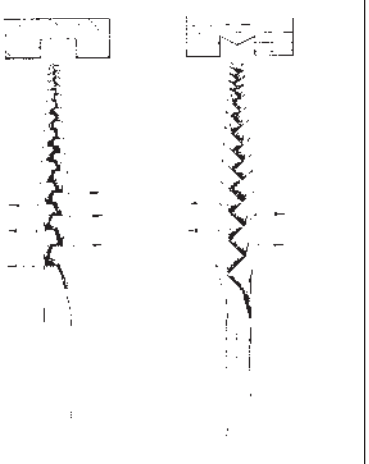
To sharpen any saw, the saw must be held firmly in a vise that secures the saw along both sides close to the teeth. This prevents any vibration and makes saw sharpening more comfortable. You can use a saw vise designed for this purpose, or you can make one from wood (3). I prefer to clamp the saw between two boards in the bench vise. This works very well.

We sharpen saws using tapered triangular files designed specifically for saw sharpening. The files come in different sizes determined by the size of the saw teeth. A large-toothed saw requires a large saw file. The smaller the saw teeth, the smaller the file. An overly large file will not work well on small teeth because the file cut is too coarse for the thin steel of the saw and the larger file has a small flat on each corner of the file that prevents the file from cutting cleanly into the bottom of such a small gullet. The same saw file is used for both crosscutting and ripcutting saws, but the saw file size is relative to the size of the saw teeth. Three sizes of saw files will accommodate almost all sizes of handsaws. A saw file costs $3-6.

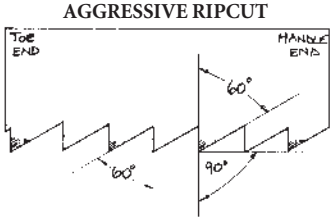
In any case, determine the size of the file by the size of the teeth: the cutting surface of the saw file should be at least twice the height of the tooth you will be filing (4). That way, when you rotate the saw file to a new corner of the file, the file will cut with fresh teeth. If the file is too small, the cutting teeth of the file will be unevenly worn on either of the two faces so when you rotate your file, one face will cut more while the other cuts less; the teeth of the saw will be cut unevenly and will vary in size. The chart below will help you determine the correct size of file to choose.

ADJUSTABLE FILE ALIGNMENT JIG

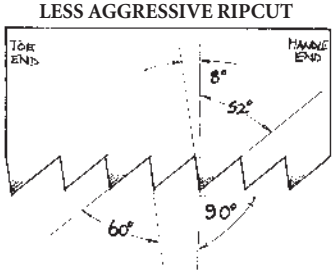
To help you establish confidence in maintaining the angle of the saw file, I suggest you make an alignment jig that will help you hold the file at a consistent pitch throughout the filing of a particular saw. This jig can be adjusted for any angle

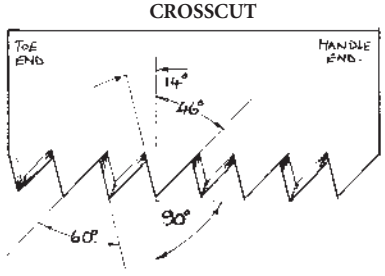


1 ARROWS INDICATE SET OF TEETH



You can vary the pitch of the teeth from 0°–8°. Because saw files have a fixed angle of 60°, whatever pitch you choose for the front (rake angle) of the teeth determines the angle of the back of the teeth: the combined angles of the front and back will always equal 60°.

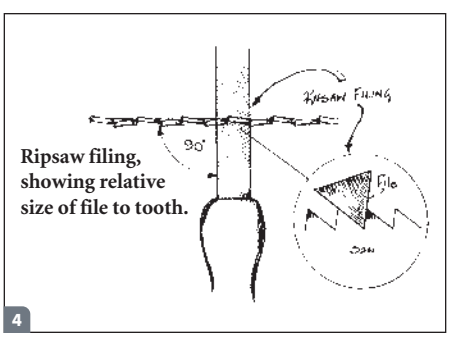




2



3



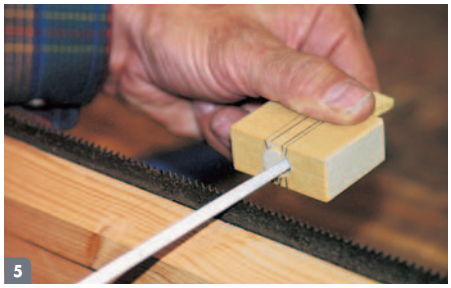
Ripsaw filing, showing relative size of file to tooth.

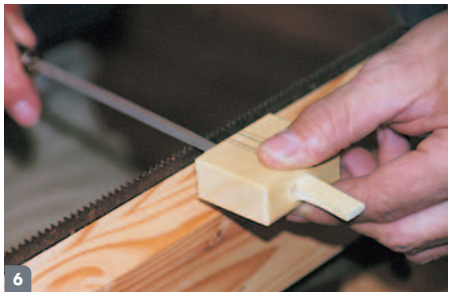
4

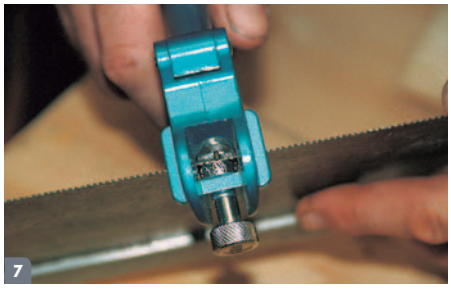
SAW FILE CHART

SIZE OF FILE	TYPE OF SAW	PPI
4" XX slim	Gents saw	16-22
6" XX slim	Dovetail saw	13-15
7" XX slim	Midsize back saws	11-14
7" X slim	Hand, Back, Tenon saw	8-10
7" slim	Handsaw	5-7

using the same hole by releasing the file and turning the dowel. You can make separate ones to accommodate a variety of pitches to suit your needs or even put two or three holes in one jig and use permanent dowels glued into the holes at various angles. The jig fits onto the tip of the saw file. By adjusting the file in the hole to the angle you want, then, holding the jig level with the tooth edge of the saw, you will be able to maintain a consistent angle as you file the saw teeth to the pitch or rake you choose (**5**,**6**).


5


6


7

TWO TYPES OF SHARPENING

The simplest saws to sharpen are ripsaws. This pattern of sharpening creates chisel-type teeth that cut a narrow kerf the width of the thickness of the saw blade plus whatever set is on the teeth. The set of the teeth widens the cut slightly to prevent the saw from binding as you cut deeper into the wood. Softwoods may require slightly more set than hardwoods but most people set their saws according to the number of points there are to the inch. Both crosscut and ripsaws require the teeth to be set. Cutting with (along) the grain gives a pleasant, smooth cutting sensation. Ripsaws cut only on the forward or push stroke, with each tooth cutting a small, flat-bottomed groove to receive the body of the saw with each subsequent, deepening cut.

The chisel-type teeth of the ripsaw are not capable of severing cross-grain fibers; instead, they tear and wrench the wood, forcing the fibers apart. Unlike ripsaws, crosscut saws have pointed knife-like teeth that slice crossways through the fibrous strands of the grain. Crosscut saws cut on both the forward and pull stroke, though generally they cut much more aggressively and remove more material on the forward stroke because of the rake angle the teeth are sharpened at. Crosscut saws usually require little downward pressure, the nature of the cut and the shape of the teeth are completely different than the rip cut. Usually the combined weight of the saw and hand provides all the pressure you need. In actuality, when you use a well-sharpened saw of any kind, you will find the saw works best when you learn to suspend and feed the saw into the fibers of the wood, especially on the opening (beginning) cuts.

SETTING SAW TEETH

We set the saw teeth by slightly bending each alternate tooth to widen the kerf as the saw passes into and through the wood. Setting each tooth this way reduces the drag on the saw and prevents it from binding in the cut. The extra width of the cut, the saw kerf, also allows for minor realignment of the saw to correct any deviation in the course of the saw as you progress through the board. The binding of the saw blade in the kerf is primarily caused by thousands of minute fibers that remain on the inside walls as the saw cuts. As the saw passes deeper into the cut, these tiny fibrous tips bend forward in the direction of the saw cut and bind against the cheeks of the saw. The wetter the wood, the more the saw will bind.

We use a unique tool called a saw set to set the teeth of the saw. These saw sets generally have a pistol-grip handle with a squeezing mechanism that forces a plunger against each tooth to bend the tooth uniformly against the anvil of the saw set (**7**). The anvil is a small adjustable wheel with numbered settings that correspond to the number of points per inch of saw length. We find the correct setting by counting the number of tooth tips or points that fall within each inch of saw length. For a six-point saw we set the dial wheel to #6. There are two saw sets available: one sets teeth from 4-12 ppi, and the other sets teeth from 13-27 ppi.

To set the teeth, choose the correct setting on the saw set and place the saw set on the tooth edge of the saw. I suggest that for your first attempt at saw setting and sharpening, you try to find a saw with larger teeth than, say, a dovetail saw; 6-8 teeth would work well.

The manufacturer will probably have already set your saw, but through continuous use, sharpening and so on, the set lessens and requires restoring. Look at the existing set so you will know which tooth to bend. You will bend the teeth that are leaning away from you. If you cannot tell by looking, try feeling the teeth. You should be able to feel which teeth are leaning away from you. If you can't, it may be that all the set is gone. If that's the case, you can begin setting any tooth; it will probably be easiest to start with the first one. If you can determine which teeth are set, place your saw set with the piston plunger of the saw set centered on the first tooth that leans away from you. Now squeeze the pistol-grip plunger mechanism. You will be able to see the tooth bend as the plunger forces the tooth against the anvil. Now skip the adjacent tooth, the one leaning towards you, and set the next tooth that leans away from you. Continue setting each alternate tooth down the full length of the saw. With this side of the saw completed, turn the saw around and set the opposite teeth. You are now ready to sharpen the saw.

Note that you can set your saw teeth either before or after sharpening your saw. You don't have to set your saw each time you sharpen, though you can if you want to. We usually determine the need for setting the saw if the saw begins to bind in the cut.

Let me say here that in most cases, saws that come from manufacturers generally have too much set on the teeth. This severely impairs the performance of the saw and you may feel that the problem lies in you. In cases like this, take the saw blade and lay it flat on a steel table; a jointer bed or tablesaw table works well. With a hammer, gently tap the teeth,

working from one end of the saw to the other. Flip the saw over and do the same to the teeth on the other side. Now try the saw to see if it cuts well. Usually this takes care of the problem, but if the saw now binds in the cut, reset the teeth with the correct setting on the saw set.

SHARPENING THE RIPSAW

With the saw held firmly, begin at either end of the saw by placing the saw file directly in the gullet of the saw. You will begin by putting the file onto the back of the first tooth that leans away from you, which will therefore automatically place the opposite side of the file on the front of the tooth leaning towards you. When you push the saw file through the gullet, you will be filing the back of one tooth and the front of the adjacent tooth in one pass with the file.

If the saw is in reasonably sharp condition, one pass may be enough. If not, take one or two more passes but no more than a total of three at this stage. If the saw needs more sharpening, it's better to initially complete both sides, taking three strokes per gullet and then begin again with one or two further passes. It's important to remember that you must try to maintain an even pressure throughout each stroke and that each file stroke is the same length. This will help to maintain a consistent tooth size and pattern. Otherwise it's easy to over-sharpen individual teeth or sections of teeth and distort them. It's also important to make sure you take equal-length strokes with consistent, equal pressure. Any excessive variation will again distort the teeth, and you will have high and low teeth. You can usually correct this discrepancy by skipping over the low cuts in subsequent sharpening until all the teeth become level. If, however, you ignore the problem and continue sharpening in the low spots, the saw will deteriorate with subsequent sharpening. As you did with the saw setting, skip over each alternate tooth until the whole side is completed. Turn the saw around and again sharpen the back of the teeth leaning away from you and the front of the teeth leaning towards you. You will automatically be sharpening the unsharpened teeth if you follow this pattern. If a saw is relatively new or in good condition, you can use the existing angle of the saw teeth as a guide for the file, otherwise you will have to correct any discrepancy. Most craftsmen who sharpen their own saws pay little attention to specific angles, but they do consider how much rake to put on the teeth and vary it according to their own preference. The rake is the forward angle of the tooth. You can sharpen to a very aggressive angle or alter the angle for a less aggressive cut, depending on the wood and the nature of the work.

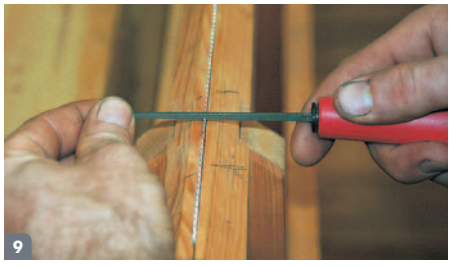
The ripsaw is the easier of the two saws to sharpen because you file the teeth at 90° to the length of the saw body and 90° to the vertical blade (**4,8**). A slight variation makes little noticeable difference to the cutting efficiency of the saw, but it's always good to strive for accuracy. With the ripsaw, you could sharpen all the teeth from one side, but the reason we cut from both sides is that each pass with the file creates a small burr on the opposite side of the teeth. If the burr is all on one side, the saw cuts unequally and deviates from the line of cut.

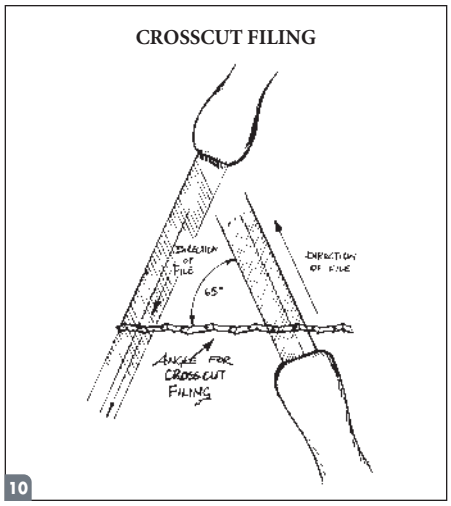
SHARPENING CROSSCUT SAWS

A crosscut saw requires a little more effort to sharpen than a ripsaw, but the difference in the cutting efficiency on cross-grain fibers is staggering.

To sharpen the crosscut saw we begin in the same way we did with the ripsaw, by determining the direction of the set. The difference is that we will not be filing at a 90° angle to the length of the saw, but at an angle of approximately 65°. However, we will still file the teeth at 90° to the vertical blade, skipping each alternate tooth and reversing the saw to file the opposite teeth from the other side. To help guide you with the angle for filing, mark the 65° angle directly onto the saw chocks or the battens that secure the saw in the vise (**9,10**). Again, remember to use consistent file strokes with even pressure,


8


9



10


11

otherwise you will distort the saw teeth. This last filing will bring all the teeth to a point because both the front and back of each tooth has a bevel. Again, we will be filing the back of the tooth leaning away from us and also the front of the tooth leaning towards us. The difference this time is important though not critical, but I suggest you follow all the recommended angles in the drawings to guide you. As you develop your skills you will be able to sharpen different saws of the same size to suit particular applications. For instance, you may sharpen one saw to cut aggressively with less of a clean cut and sharpen another for a less aggressive cut with a focus on accuracy. For some saws you will want less set than others, and in some cases, you may want no set at all.

JOINTING OR TOPPING THE TEETH

After several years of sharpening, saws may become hollow or bowed along the edge, or the teeth may develop irregularities that cannot be corrected by reshaping the individual teeth. To restore the teeth to full height and straighten the saw along its cutting edge, we use a simple procedure we call jointing or topping. We generally joint the saw before any general sharpening takes place. We use a smooth-cut mill file to joint the teeth level with each other. Remove the handle from the file. Begin with the file at the handle end of the saw pointing towards the saw tip. Lay the file flat on the teeth and straddle the file with the palm of your right hand, using your fingertips against each side of the saw to center the file and guide it along the tops of the saw teeth (**11**). Place the heel of your left hand on the front end of the file and, with an even, light pressure, push the file carefully along the top of the teeth. Take a look at the tips of the teeth to ensure that tiny flats occur at the top of all the teeth. You may find gaps of several teeth that have no flat tops or just an occasional tooth. If this happens, repeat the filing until all the teeth have flat spots. Occasionally, saws lose a tooth for various reasons. It's not necessary to joint all the teeth of the saw down to salvage one or two teeth. Subsequent sharpening will restore the tooth over a period of time.

Paul Sellers has been a woodworker for 40 years. He teaches at Homestead Heritage School of Woodworking in Waco, Texas.

Sharpening New Handsaws

IT CAN BE DISAPPOINTING to buy a new handsaw direct from the manufacturer and find it hardly cuts through even the softest wood. Before you send it back, however, remember that you ordered the saw because you needed one. With a little work, you can take such a saw and make it cut like any of the finest saws ever produced.

I have used many of my saws now for over 30 years. Recently, I realized that all those years of sharpening had reduced the cutting steel to the point at which they might wear out too much to be of use to my children, who also enjoy woodworking with hand tools. I decided to introduce new saws alongside my old ones; I chose saws that I felt met my standards from the Highland Hardware catalog, and first ordered an 8" straight-handled dovetail saw produced by the Crown company of England. I considered the descriptions carefully, and was surprised that they could offer such a quality saw with a brass back and solid rosewood handle for only $12.95.



When the new saw arrived, I was disappointed. All the brass was poorly finished with no polish at all. I wondered if the poor finish would also reflect the quality of the saw's cutting ability. Testing the saw on some white pine, the saw cut poorly even though, to my finger tips, the saw felt reasonably sharp. Now, if I was a novice woodworker, having nothing to compare my experience with, I would do one of two things: I would put aside the saw and say to myself it's no wonder people moved to machines, or, I would use it in ignorance, believing that this was the standard. How would I know otherwise?

When I examined the handsaw more closely, I was actually impressed with the quality of the steel blade, the solid brass back (aside from the poor finish), the beautiful rosewood handle and the brass ferule, all of which felt good. It shouldn't be necessary, but I could see that with a few minutes' work, this saw had the potential to be a truly fine handsaw. If woodworkers regarded new saws more like an unfinished kit, they could have excellent tools that would function perfectly without spending hundreds of dollars. I set to work.

To polish the brass parts, I took 320-grit paper and sanded along the length of the back to remove the surface pitting and scratches. I then worked progressively through various finer grits to 2000-grit, and finally polished the back with buffing compound on a leather strop. The saw began to look and feel like those old saws you now only see on tool calendars.

The next step was sharpening. Such a small saw might not be my first recommendation for anyone to practice on, but if you carefully follow the steps outlined in the article, you will minimize the risk of ruining your saw. The main difficulty inexperienced people might find is that they take off too much steel from the "v" of the teeth. If you take off too much steel in the first pass with the saw file on such small teeth, when you pass back from the opposite side you can slip from the tiny throat or gullet with the saw file and remove the whole tooth in one stroke, leaving a deep gullet in the midst of all the smaller teeth. This comes from applying excessive pressure on the file. On such fine-toothed saws with teeth already close to sharp, you need only a lightweight pass with the file.

Another fault with almost every direct-from-the-factory saw is too much set. The right amount of set will vary according to practical or personal preference. You can set a saw's teeth minimally for some applications and aggressively for others. Follow the guidelines in the article and you will have a fine handsaw that will serve you for years to come.

—Paul Sellers